

CHAPTER 1

INTRODUCTION

In response to the growing concerns surrounding women's safety globally, this project introduces a groundbreaking security solution – a patrolling robot equipped with an ESP32CAM. This innovative robot is designed for deployment in both secluded areas and public spaces, serving as a vigilant guardian against potential threats. The heart of the system lies in its ESP32CAM, a versatile and powerful microcontroller with an integrated camera module.

The robot operates on a mobile platform, allowing it to navigate diverse terrains and access hard-to-reach areas. The ESP32CAM facilitates continuous surveillance and real-time monitoring of the designated premises, ensuring a proactive response to any security issues. This cutting-edge technology empowers law enforcement and security personnel to address safety concerns promptly.

The comprehensive security solution aims to contribute to the overall well-being and security of individuals, with a particular focus on the safety of women in various environments. By leveraging advanced robotics and surveillance capabilities, the project provides a technologically advanced approach to enhance public safety. The deployment of these patrolling robots in strategic locations can act as a deterrent to potential threats and promote a safer environment for everyone. This initiative signifies a commitment to leveraging technology for the greater good and addressing the urgent need for enhanced security measures in today's complex societal landscape.

Using an ESP32 for a women's safety patrolling robot is a great idea! The ESP32's capabilities, such as wireless connectivity and low power consumption, can be leveraged for features like remote monitoring and sending alerts to authorities or loved ones.

Overall, the women safety patrolling robot represents a powerful tool in the ongoing effort to ensure the safety and well-being of women. by harnessing the capabilities of the esp32 microcontroller, it offers a versatile and effective solution for addressing safety concerns in various situations

1.1 OBJECTIVE

The objective of this project is to address the pressing concern of women's safety in isolated areas by implementing a security patrolling robot utilizing ESP32CAM technology. The robotic system, equipped with a camera, aims to monitor and secure premises in real-time, providing live streaming and continuous surveillance. Controlled remotely via mobile or laptop using an IP address, the robot allows directional commands for navigation and incorporates an alert system through a buzzer, enhancing safety measures and contributing to a secure environment. By achieving these objectives, the Women Safety Patrolling Robot project aims to provide women with a reliable and proactive safety solution that can help mitigate risks and enhance their sense of security in public spaces.

CHAPTER 2

LITERATURE SURVEY

2.1 A Mobile Based Women Safety Application (I Safe Apps)

Description: Many unfortunate incidents have been taking place in woman's case. Problems may come from any direction such as women walking on the road after the work, going to super market or many other reasons for which they go alone. People at home are not sure of their return safely. Another factor is woman die without knowing the reason as they attend excursions and industrial trips conducted by the organizations. It happens due to attacks on woman but not suicides. In 2013 there happened an incident which is a gag rape in New Delhi in the case of 23 year old woman in bus at 9:30 PM. Another incident that has taken place at Mumbai in the case of woman who is leaving her native place after Christmas holidays has been kidnapped and killed. These are some of the problems that have taken place in the day to day life of women. In order to overcome such problems faced by women the I Safety (women security apps) mobile based application is not only necessary to use but also plays a pivotal role with android software.

2.2 Abhaya: An Android App for the safety of women

Description: In today's world, people using smart phones have increased rapidly and hence, a smart phone can be used efficiently for personal security or various other protection purposes. The heinous incident that outraged the entire nation have waken us to go for the safety issues and so a host of new apps have been developed to provide security systems to women via their phones. This paper presents Abhaya, an Android Application for the Safety of Women and this app can be activated this app by a single click, whenever need arises. A single click on this app identifies the location of place through GPS and sends a message comprising this location URL to the registered contacts and also call on the first registered contact to help the one in dangerous situations. The unique feature of this application is to send the message to the registered contacts continuously for every five minutes until the "stop" button in the application is clicked. Continuous location

tracking information via SMS helps to find the location of the victim quickly and can be rescued safely.

2.3 Violence Against Homeless Women: Safety and Social Policy

Description: Over the past 20 years there has been increasing understanding of the gendered nature of homelessness in Australia. Most significantly, this gendering has occurred through the acknowledgement of the links between domestic violence and homelessness and this has played out in Australian social policy through the funding of specialist domestic violence services. However, not all women are assisted by these specialist services—either because they are not homeless due to domestic violence, or because they fall through the gaps in the service system. Homelessness exposes these women to heightened vulnerability to violence. This article considers homeless women's experiences of violence and their implications for homelessness policy. Framed by Australian and Victorian social policy and drawing on a qualitative study of 29 women, all of whom had experienced violence during homelessness, the article argues that greater policy attention needs to be paid to ensuring homeless women's safety.

2.4 Smart Wearable Device for Women Safety Using IoT

Description: The crimes against women have been rising significantly and often hear about molestation, eve-teasing and rape cases in the public places of the society. The security of women is the most important concern these days and to build a safety device to act as a rescue and to prevent from harm at the time of hazard is highly necessary especially for women. In this paper, a smart device for women's safety which automates the emergency alert system by using pressure sensor, pulse-rate sensor and temperature sensor to detect a possible atrocity automatically using outlier detection is proposed. This system detects and sends the alerts for the dear ones with the

CHAPTER 3

EXISTING SYSTEM

The existing system for women's safety lacks a robust and real-time response mechanism. Introducing a Women Safety Robot utilizing RF (Radio Frequency) technology addresses this gap. This innovative robot incorporates RF sensors to detect distress signals from women in need, sending immediate alerts to a centralized monitoring system. The RF system enables precise location tracking, facilitating a quicker and more effective response from authorities. The robot is equipped with communication capabilities, offering a direct link between the victim and emergency services. This integration enhances women's safety by providing a proactive and technologically advanced solution, contributing to a more secure and responsive environment for women in various settings.

3.1 DRAWBACKS

1. Existing system lacks real-time response, delaying assistance to women in distress.
2. RF technology focuses on distress signals, missing proactive surveillance features.
3. RF system lacks direct victim-emergency service link, hindering quick and effective communication.
4. The existing system may face limitations in precision and efficiency compared to proposed Wi-Fi-enabled robot.

CHAPTER 4

PROPOSED SYSTEM

The robotic vehicle is constructed with a robust chassis, specifically designed to accommodate the ESP32CAM module. The ESP32CAM, equipped with a high-resolution camera, is securely mounted on the robot to capture clear and detailed images.

For wireless communication, the robot incorporates Wi-Fi protocols, enabling remote control and monitoring. A unique IP address is assigned to the robot, facilitating seamless control through a mobile device or laptop.

To enhance security capabilities, a buzzer is integrated to generate alert signals when required. The ESP32CAM is programmed to capture and transmit live streaming video, facilitated by the implemented Wi-Fi module. Sophisticated image processing algorithms are employed to monitor each area for potential threats or unusual activities, ensuring a proactive surveillance system.

This comprehensive hardware and software setup creates a versatile robotic surveillance vehicle capable of remote operation, live video streaming, and efficient threat detection through advanced image processing.

4.1 BENEFITS:

1. The robot provides an additional layer of security, especially during nighttime, offering a constant presence and surveillance in areas of concern.
2. Live streaming capabilities allow for real-time monitoring, enabling prompt responses to potential threats or suspicious activities.
3. Users can control and monitor the robot remotely using a mobile device or laptop, allowing for flexible and convenient surveillance.
4. By deploying a robotic solution, the risk to human patrollers, especially during night shifts, is significantly reduced, promoting a safer approach to security.

4.2 PROPOSED BLOCK DIAGRAM

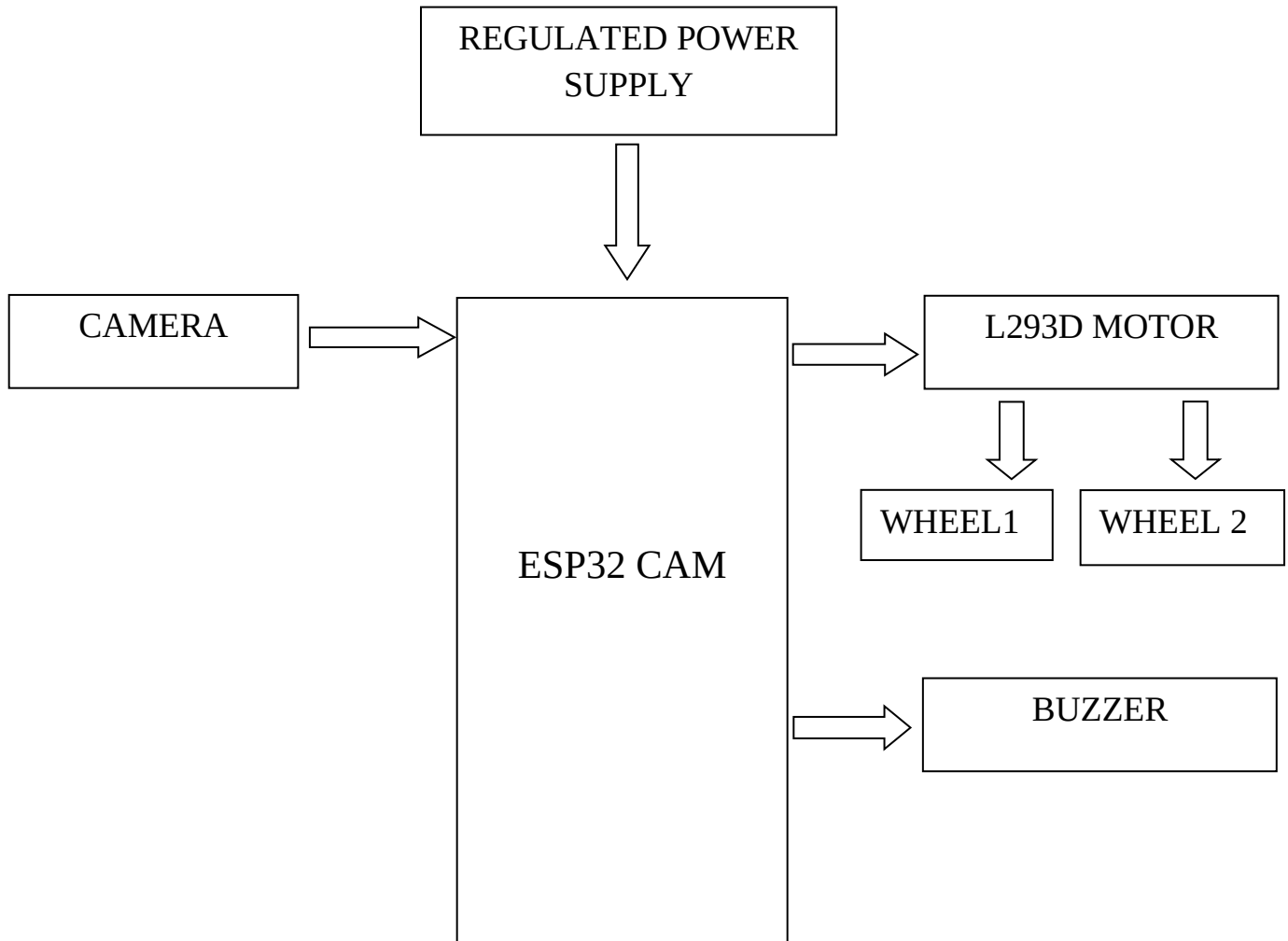


Fig 4.1: Block Diagram of Women Safety Night Patrolling Robot

4.3 HARDWARE TOOLS

- ESP32 CAM MODULE
- L293D DRIVER
- MOTORS
- CAMERA
- BUZZER
- REGULATED POWER SUPPLY

DESCRIPTION OF TOOLS

4.3.1 ESP32 camera module:

The ESP32-CAM is a very small camera module with the ESP32-S chip that costs approximately \$10. Besides the OV2640 camera, and several GPIOs to connect peripherals, it also features a microSD card slot that can be useful to store images taken with the camera or to store files to serve to clients.

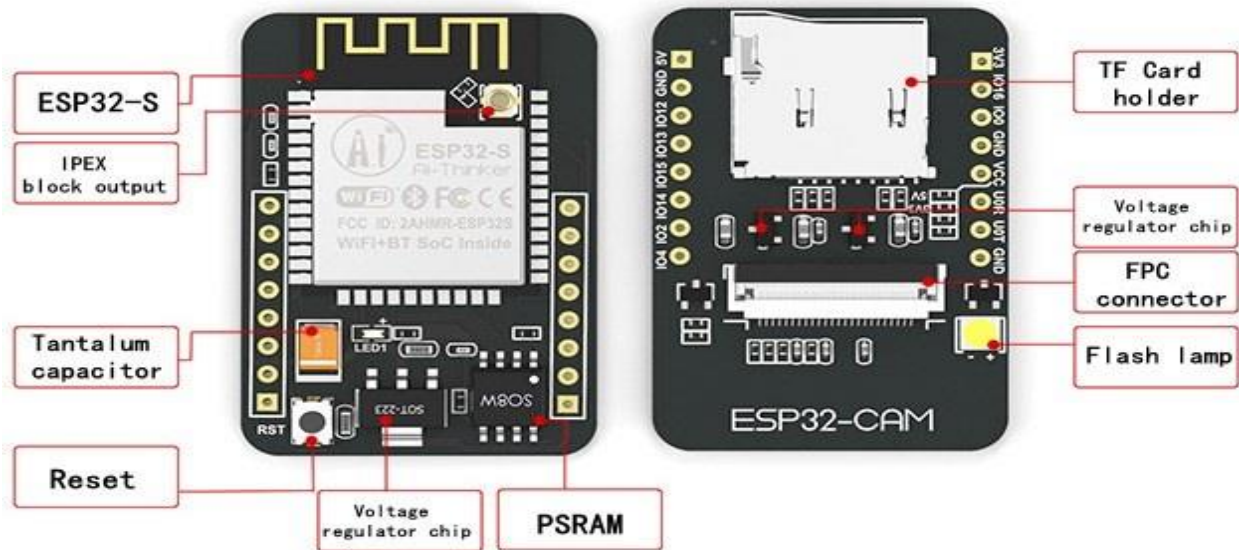


Fig 4.2: Block Diagram of ESP32

Features

Here is a list with the ESP32-CAM features:

- The smallest 802.11b/g/n Wi-Fi BT SoC module
- Low power 32-bit CPU, can also serve the application processor
- Up to 160MHz clock speed, summary computing power up to 600 DMIPS
- Built-in 520 KB SRAM, external 4MPSRAM

- Supports UART/SPI/I2C/PWM/ADC/DAC
- Support OV2640 and OV7670 cameras, built-in flash lamp
- Support image WiFi upload
- Support TF card
- Supports multiple sleep modes
- Embedded Lwip and FreeRTOS
- Supports STA/AP/STA+AP operation mode
- Support Smart Config/AirKiss technology
- Support for serial port local and remote firmware upgrades (FOT)

4.3.2 L293D

Concept

The L293D is a popular motor driver IC that can control two DC motors or one stepper motor. It's commonly used in robotics and other projects to control motor direction and speed using microcontrollers like Arduino.

It works on the concept of H-bridge. H-bridge is a circuit which allows the voltage to be flown in either direction. As you know voltage need to change its direction for being able to rotate the motor in clockwise or anticlockwise direction, Hence H-bridge IC are ideal for driving a DC motor.

There are two Enable pins on l293d. Pin 1 and pin 9, for being able to drive the motor, the pin 1 and 9 need to be high. For driving the motor with left H-bridge you need to enable pin 1 to high. And for right H-Bridge you need to make the pin 9 to high. If anyone of the either pin1 or pin9 goes low then the motor in the corresponding section will suspend working. It's like a switch.

The L293D is assembled in a 16-lead plastic package which has 4 center pins connected together and used for heat sinking. The L293DD is assembled in a 20-lead surface mount which has 8 center pins connected together and used for heat sinking.

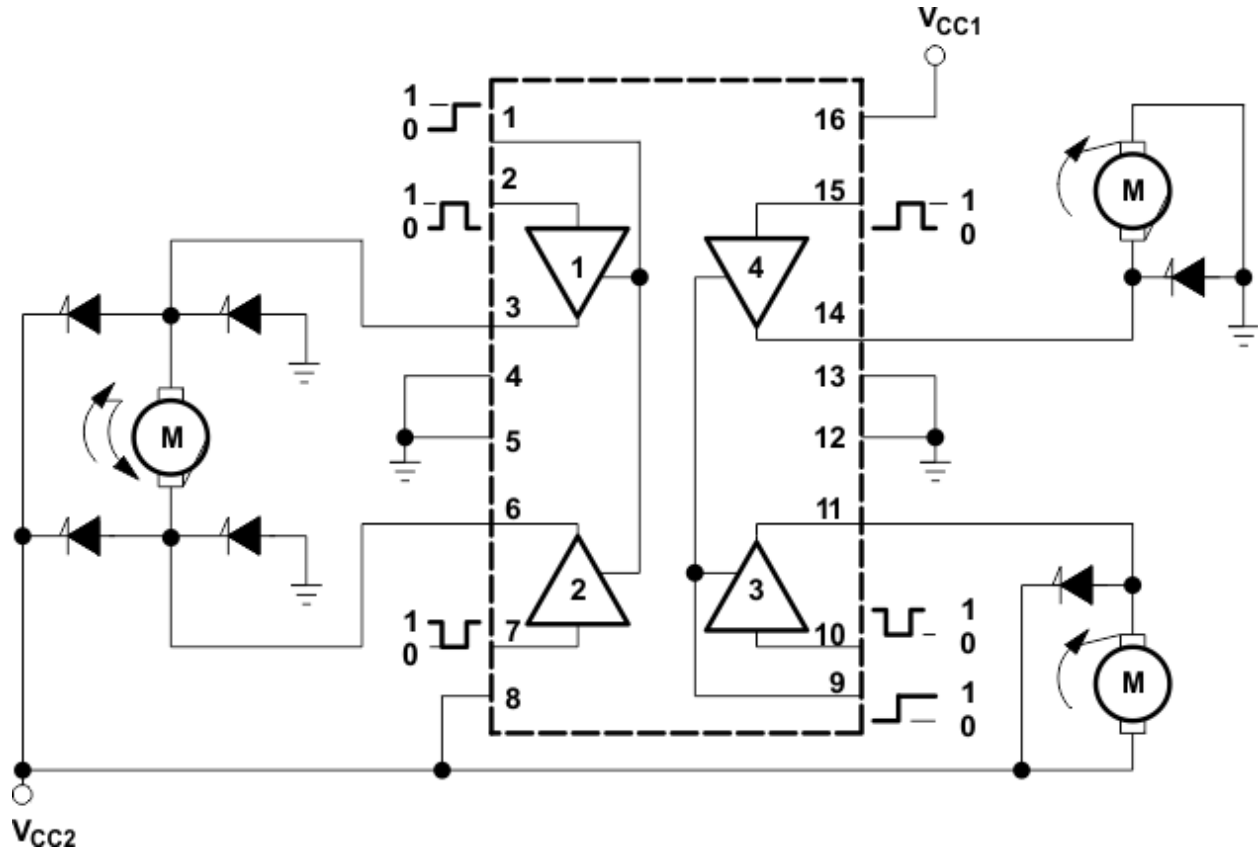


Fig 4.3: Block Diagram of L293D

In a single L293D chip there are two h-Bridge circuit inside the IC which can rotate two d c motor independently. Due its size it is very much used in robotic application for controlling DC motors. Given below is the pin diagram of a L293D motor controller.

There are two Enable pins on l293d. Pin 1 and pin 9, for being able to drive the motor, the pin 1 and 9 need to be high. For driving the motor with left H-bridge you need to enable pin 1 to high. And for right H-Bridge you need to make the pin 9 to high. If anyone of the either pin1 or p in9 goes low then the motor in the corresponding section will suspend working. It's like a switch.

TIP: you can simply connect the pin16 VCC (5v) to pin 1 and pin 9 to make them high.

The Device is a monolithic integrated high voltage, high current four channel driver desgined to accept standard DTL or TTL logic levels and drive inductive loads (such as relays solenoids, DC and stepping motors) and switching power transistors. To simplify use as two bridges eac

h pair of channels is equipped with an enable input. A separate supply input is provided for the logic, allowing operation at a lower voltage and internal clamp diodes are included. This device is suitable for use in switching applications at frequencies up to 5 kHz.

Working of L293D

There are 4 input pins for l293d, pin 2,7 on the left and pin 15 ,10 on the right as shown on the pin diagram. Left input pins will regulate the rotation of motor connected across left side and right input for motor on the right-hand side. The motors are rotated on the basis of the inputs provided across the input pins as LOGIC 0 or LOGIC 1.

In simple you need to provide Logic 0 or 1 across the input pins for rotating the motor.

L293D Logic

Let's consider a Motor connected on left side output pins (pin 3,6). For rotating the motor in clockwise direction, the input pins have to be provided with Logic 1 and Logic 0.

- **Pin 2 = Logic 1** and **Pin 7 = Logic 0** | Clockwise Direction
- **Pin 2 = Logic 0** and **Pin 7 = Logic 1** | Anticlockwise Direction
- **Pin 2 = Logic 0** and **Pin 7 = Logic 0** | Idle [No rotation] [Hi-Impedance state]
- **Pin 2 = Logic 1** and **Pin 7 = Logic 1** | Idle [No rotation]

In a very similar way, the motor can also operate across input pin 15,10 for motor on the right-hand side.

4.3.3 MOTORS

A DC motor is defined as a class of electrical motors that convert direct current electrical energy into mechanical energy. A DC motor is an electrical machine that converts electrical energy into mechanical energy. In a DC motor, the input electrical energy is the direct current which is transformed into the mechanical rotation. Here we are using the DC motors. Brushed DC electric motor generates torque directly from DC power supplied to the motor by using internal commutation, stationary magnets, and rotating electromagnets. sometimes with added dispersed copper to improve conductivity. Very small brushes may rely on sliding contact with a metal brush holder to carry current into the brush, or may rely on a contact spring pressing on the end of the

brush. The brushes in very small, short-lived motors, such as are used in toys, may be made of a folded strip of metal that contacts the commutator.



Fig 4.4: DC Motor

A DC motor's speed can be controlled over a wide range, using either a variable supply voltage or by changing the strength of current in its field windings. Larger DC motors are currently used in propulsion of electric vehicles, elevator and hoists, and in drives for steel rolling mills.

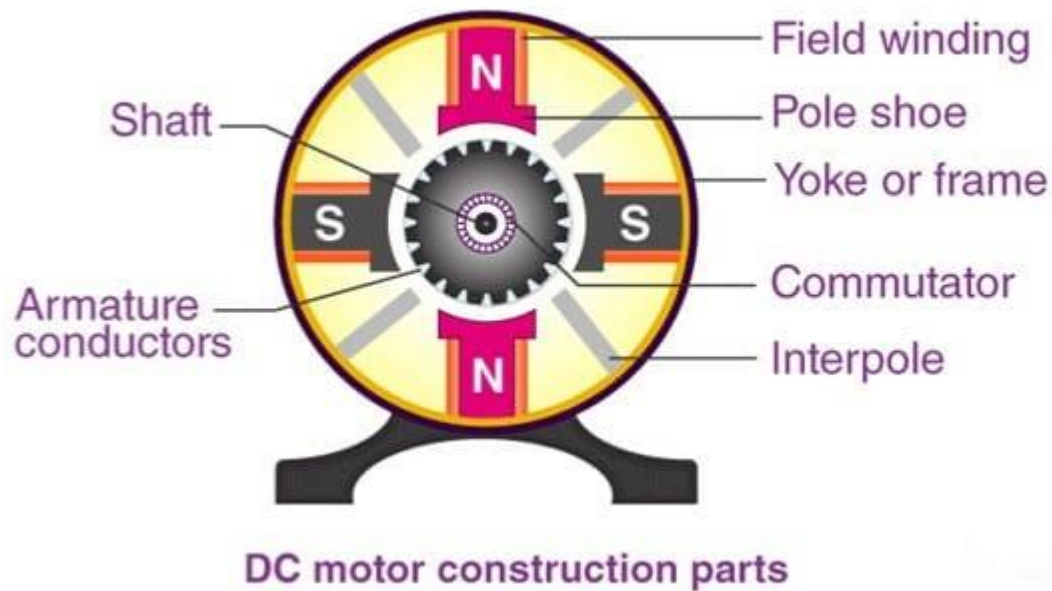


Fig 4.5: Construction of DC Motor

WORKING

Construction A magnetic field arises in the air gap when the field coil of the DC motor is energized. The created magnetic field is in the direction of the radii of the armature. The magnetic

field enters the armature from the North pole side of the field coil and “exits” the armature from the field coil’s South pole side. The conductors located on the other pole are subjected to a force of the same intensity but in the opposite direction. These two opposing forces create a torque that causes the motor armature to rotate.

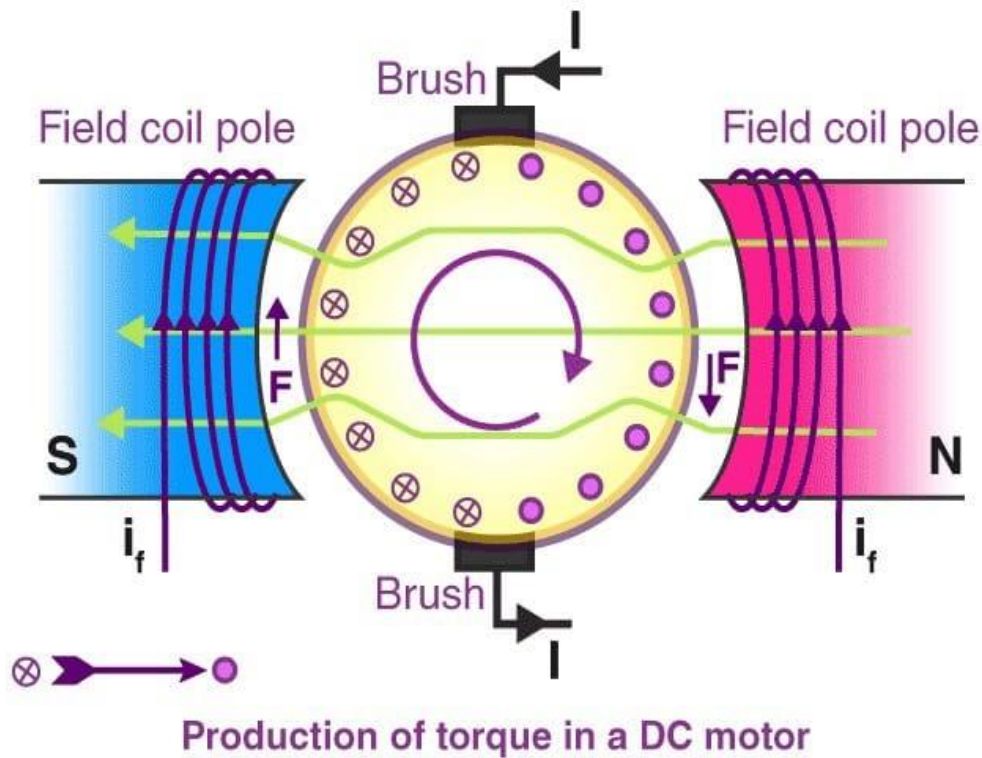


Fig 4.6: Working of DC Motor

4.3.4 CAMERA

A camera is an optical instrument that captures images. Most cameras can capture 2D images, while some more advanced models can capture 3D images. At a basic level, most cameras consist of a sealed box (the camera body), with a small hole (the aperture) that allows light to pass through and capture an image on a light-sensitive surface (usually a digital sensor or photographic film). Cameras have various mechanisms to control how light falls onto the light-sensitive surface, including lenses that focus the light and a shutter that determines the amount of time the photosensitive surface is exposed to the light. The still image camera is a key instrument in the

art of photography. Captured images may be reproduced later through processes such digital imaging or photographic printing. Similar artistic fields in the moving-image camera domain include film, videography, and cinematography.



Fig 4.7: Camera

- The word *camera* comes from *camera obscura*, which is Latin for "dark chamber" and refers to the original device used to project a 2D image onto a flat surface. The modern photographic camera evolved from the camera obscura.

4.3.5 BUZZER:

A **buzzer** or **beeper** is an audio signaling device, which may be mechanical, electromechanical, or electronic. Typical uses of buzzers and beepers include alarms, timers and confirmation of use. In a magnetic buzzer, a current is driven through a coil of wire which produces a magnetic field. A flexible ferromagnetic disk is attracted to the coil when the current is present and returns to a "rest" position when the current is not flowing through the coil, input such as a mouse click or keystroke.

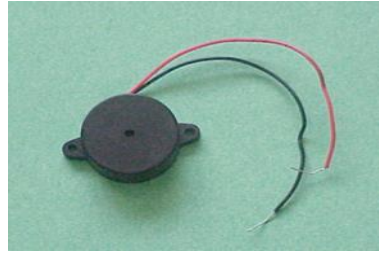


Fig 4.8: Buzzer

FEATURES

- They exhibit extremely low power consumption in comparison to electromagnetic units.
- They are constructed without switching contacts to ensure long life and no electrical noise.
- Compact, yet produces high acoustic output with minimal voltage.

Mechanical

A joy buzzer is an example of a purely mechanical buzzer.

Electromechanical

Early devices were based on an electromechanical system identical to an electric bell without the metal gong. Similarly, a relay may be connected to interrupt its own actuating current, causing the contacts to buzz. Often these units were anchored to a wall or ceiling to use it as a sounding board. The word "buzzer" comes from the rasping noise that electromechanical buzzers made.

VOLTAGE BUZZER SOUND CONTROLS

When resistance is connected in series (as shown in illustrations (a) and (b)), abnormal oscillation may occur when adjusting the sound volume. In this case, insert a capacitor in parallel to the voltage oscillation board (as shown in illustration (c)). By doing so, abnormal oscillation can be prevented by grounding one side. However, the voltage VB added to the voltage oscillation

board must be within the maximum input voltage range, and as capacitance of $3.3\mu\text{F}$ or greater should be connected.

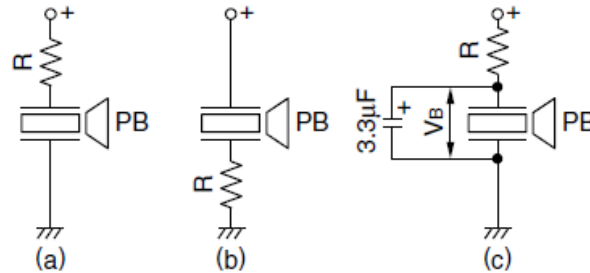


Fig 4.9: Voltage Buzzer Controls

4.3.6 REGULATED POWER SUPPLY:

All digital circuits require regulated power supply. In this article we are going to learn how to get a regulated positive supply from the mains supply.

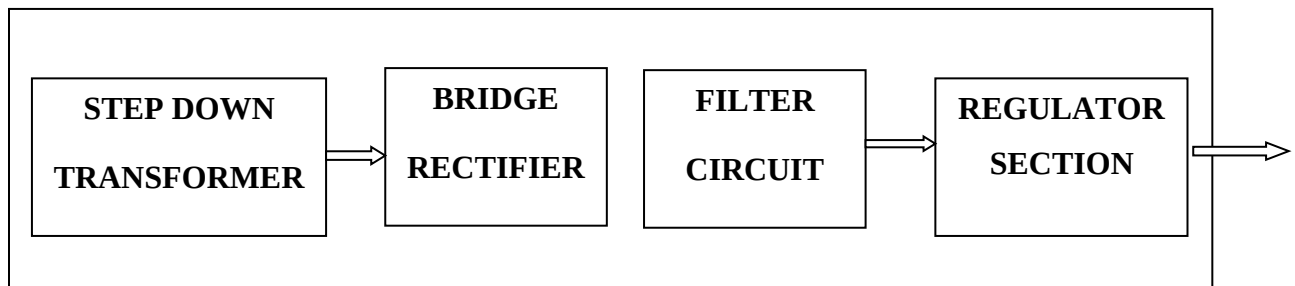


Fig 4.10: Block Diagram of Regulated Power Supply

A regulated power supply is an embedded circuit; it converts unregulated AC (alternating current) into a constant DC. With the help of a rectifier, it converts AC supply into DC. Its function is to supply a stable voltage (or less often current), to a circuit or device that must be operated within certain power supply limits.

The above block diagram shows the Regulated Power Supply and its components. In these various components are presented to work in the form of RPS.

TRANSFORMER

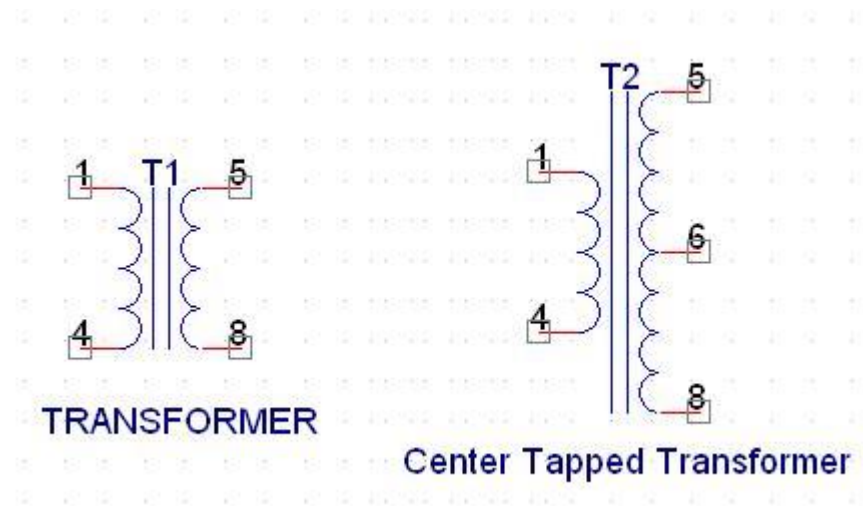


Fig 4.11 : Types Of Transformers

A transformer consists of two coils also called as “WINDINGS” namely PRIMARY & SECONDARY.

They are linked together through inductively coupled electrical conductors also called as CORE. A changing current in the primary causes a change in the Magnetic Field in the core & this in turn induces an alternating voltage in the secondary coil. If load is applied to the secondary then an alternating current will flow through the load. If we consider an ideal condition then all the energy from the primary circuit will be transferred to the secondary circuit through the magnetic field.

$$P_{\text{primary}} = P_{\text{secondary}}$$

So

$$I_p V_p = I_s V_s$$

The secondary voltage of the transformer depends on the number of turns in the Primary as well as in the secondary.

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

RECTIFIER

A rectifier is a device that converts an AC signal into DC signal. For rectification purpose we use a diode, a diode is a device that allows current to pass only in one direction i.e. when the anode of the diode is positive with respect to the cathode also called as forward biased condition & blocks current in the reversed biased condition.

Rectifier can be classified as follows:

1) Half Wave rectifier.

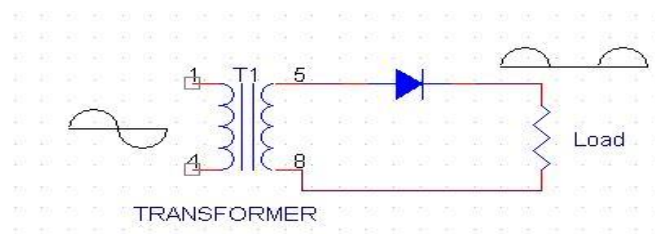


Fig 4.12: Half Wave Rectifier

This is the simplest type of rectifier as you can see in the diagram a half wave rectifier consists of only one diode. When an AC signal is applied to it during the positive half cycle the diode is forward biased & current flows through it. But during the negative half cycle diode is reverse biased & no current flows through it. Since only one half of the input reaches the output, it is very inefficient to be used in power supplies.

2) Full wave rectifier:

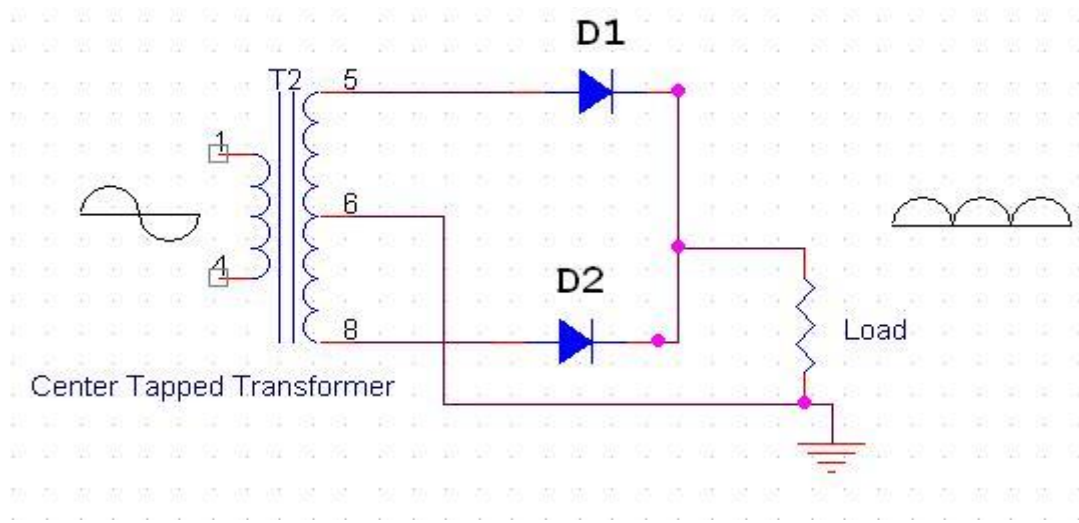


Fig 4.13: Full Wave Rectifier

Half wave rectifier is quite simple but it is very inefficient, for greater efficiency we would like to use both the half cycles of the AC signal. This can be achieved by using a center tapped transformer i.e. we would have to double the size of secondary winding & provide connection to the center. So during the positive half cycle diode D1 conducts & D2 is in reverse biased condition. During the negative half cycle diode D2 conducts & D1 is reverse biased. Thus we get both the half cycles across the load.

One of the disadvantages of Full Wave Rectifier design is the necessity of using a center tapped transformer, thus increasing the size & cost of the circuit. This can be avoided by using the Full Wave Bridge Rectifier

Full wave rectifier is defined as a rectifier that converts the complete cycle of alternating current into pulsating DC. Unlike halfwave rectifiers that utilize only the halfwave of the input AC cycle, full wave rectifiers utilize the full cycle.

3) Bridge Rectifier:

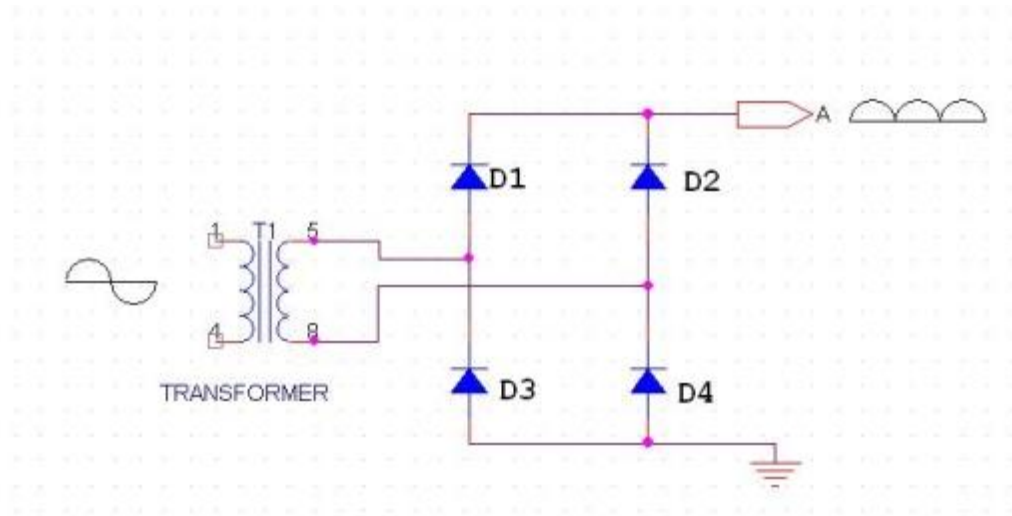


Fig 4.14: Bridge Rectifier

As the name suggests it converts the full wave i.e. both the positive & the negative half cycle into DC thus it is much more efficient than Half Wave Rectifier & that too without using a center tapped transformer thus much more cost effective than Full Wave Rectifier.

Full Bridge Wave Rectifier consists of four diodes namely D1, D2, D3 and D4. During the positive half cycle diodes D1 & D4 conduct whereas in the negative half cycle diodes D2 & D3 conduct thus the diodes keep switching the transformer connections so we get positive half cycles in the output.

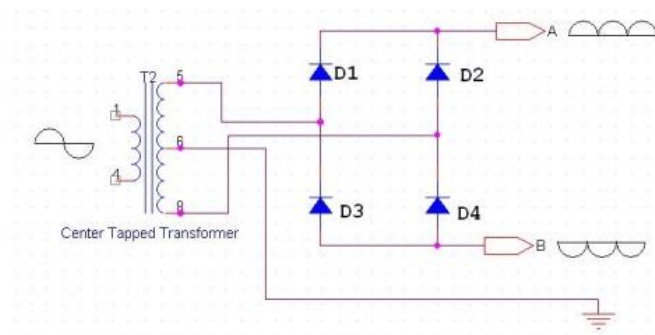


Fig 4.15: Bridge Rectifier with Center Tapped

If we use a center tapped transformer for a bridge rectifier we can get both positive & negative half cycles which can thus be used for generating fixed positive & fixed negative voltages.

FILTER CAPACITOR

Even though half wave & full wave rectifier give DC output, none of them provides a constant output voltage. For this we require to smoothen the waveform received from the rectifier. This can be done by using a capacitor at the output of the rectifier this capacitor is also called as “FILTER CAPACITOR” or “SMOOTHING CAPACITOR” or “RESERVOIR CAPACITOR”. Even after using this capacitor a small amount of ripple will remain.

We place the Filter Capacitor at the output of the rectifier the capacitor will charge to the peak voltage during each half cycle then will discharge its stored energy slowly through the load while the rectified voltage drops to zero, thus trying to keep the voltage as constant as possible.

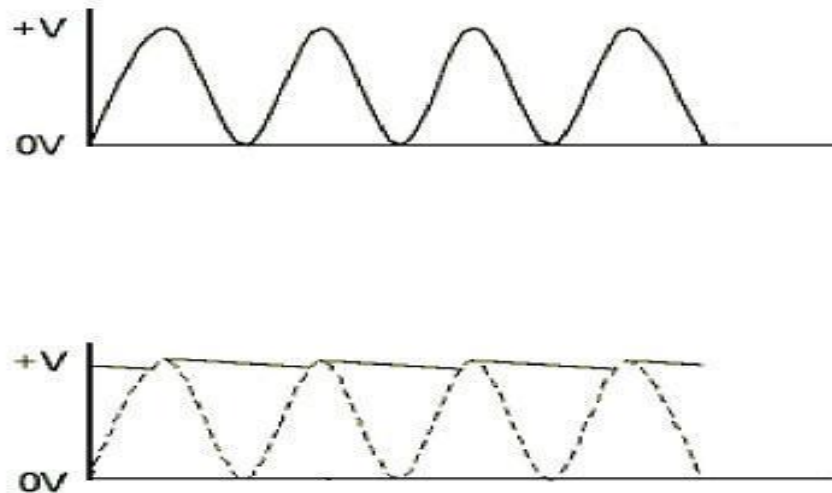


Fig.4.16: Output Waveforms

If we go on increasing the value of the filter capacitor then the Ripple will decrease. But then the costing will increase. The value of the Filter capacitor depends on the current consumed by the circuit, the frequency of the waveform & the accepted ripple.

$$C = \frac{V_r F}{I}$$

Where,

V_r = accepted ripple voltage.(should not be more than 10% of the voltage)

I = current consumed by the circuit in Amperes.

F = frequency of the waveform. A half wave rectifier has only one peak in one cycle so $F=25\text{hz}$

Whereas a full wave rectifier has Two peaks in one cycle so $F=100\text{hz}$.

VOLTAGE REGULATOR

A Voltage regulator is a device which converts varying input voltage into a constant regulated output voltage. Voltage regulator can be of two types

- 1) Linear Voltage Regulator: Also called as Resistive Voltage regulator because they dissipate the excessive voltage resistively as heat.
- 2) Switching Regulators: They regulate the output voltage by switching the Current ON/OFF very rapidly. Since their output is either ON or OFF it dissipates very low power thus achieving higher efficiency as compared to linear voltage regulators. But they are more complex & generate high noise due to their switching action. For low level of output power switching regulators tend to be costly but for higher output wattage they are much cheaper than linear regulators.

The most commonly available Linear Positive Voltage Regulators are the 78XX series where the XX indicates the output voltage. And 79XX series is for Negative Voltage Regulators.

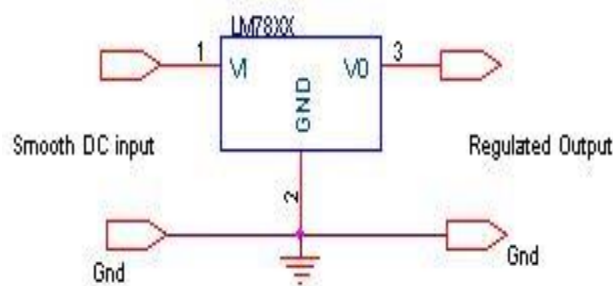


Fig 4.17: Switching Regulator

After filtering the rectifier output the signal is given to a voltage regulator. The maximum input voltage that can be applied at the input is 35V. Normally there is a 2-3 Volts drop across the regulator so the input voltage should be at least 2-3 Volts higher than the output voltage. If the input voltage gets below the V_{min} of the regulator due to the ripple voltage or due to any other reason the voltage regulator will not be able to produce the correct regulated voltage.

Power Supply

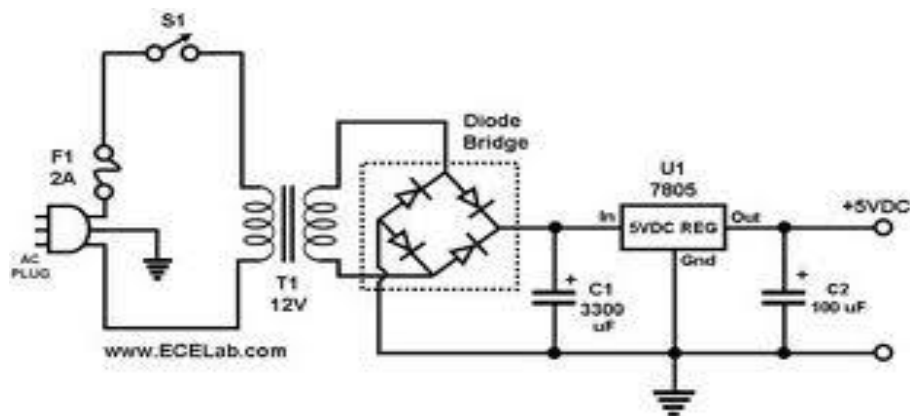


Fig 4.18: Circuit Diagram of Power Supply

IC 7805:

7805 is an integrated three-terminal positive fixed linear voltage regulator. It supports an input voltage of 10 volts to 35 volts and output voltage of 5 volts. It has a current rating of 1 amp although lower current models are available. Its output voltage is fixed at 5.0V. The 7805 also has a built-in current limiter as a safety feature. 7805 is manufactured by many companies, including National Semiconductors and Fairchild Semiconductors.

The 7805 will automatically reduce output current if it gets too hot. The last two digits represent the voltage; for instance, the 7812 is a 12-volt regulator. The 78xx series of regulators is designed to work in complement with the 79xx series of negative voltage regulators in systems that provide both positive and negative regulated voltages, since the 78xx series can't regulate negative voltages in such a system. The 7805 & 78 is one of the most common and well-known of the 78xx series regulators, as it's small component count and medium-power regulated 5V make it useful for powering TTL devices.

Table Specifications of IC7805

SPECIFICATIONS	IC 7805
V_{out}	5V
$V_{ein} - V_{out}$ Difference	5V - 20V
Operation Ambient Temp	0 - 125°C
Output I_{max}	1A

CHAPTER 5

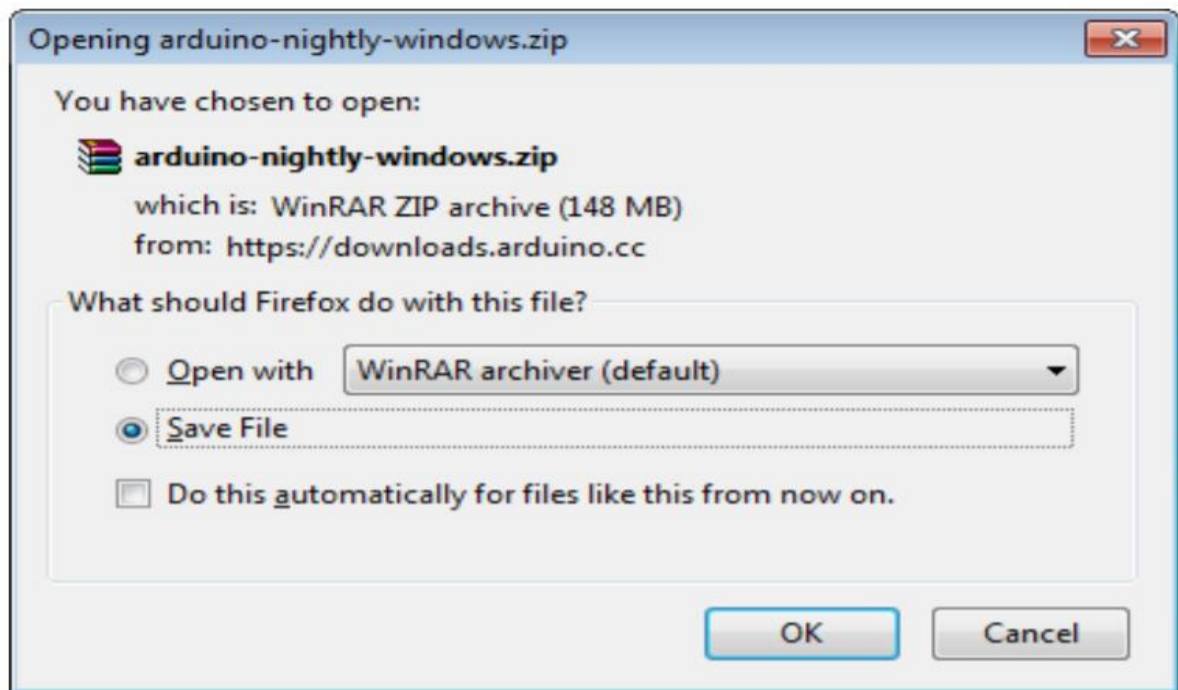
SOFTWARE TOOLS

- Arduino ide

5.1 DESCRIPTION OF TOOLS

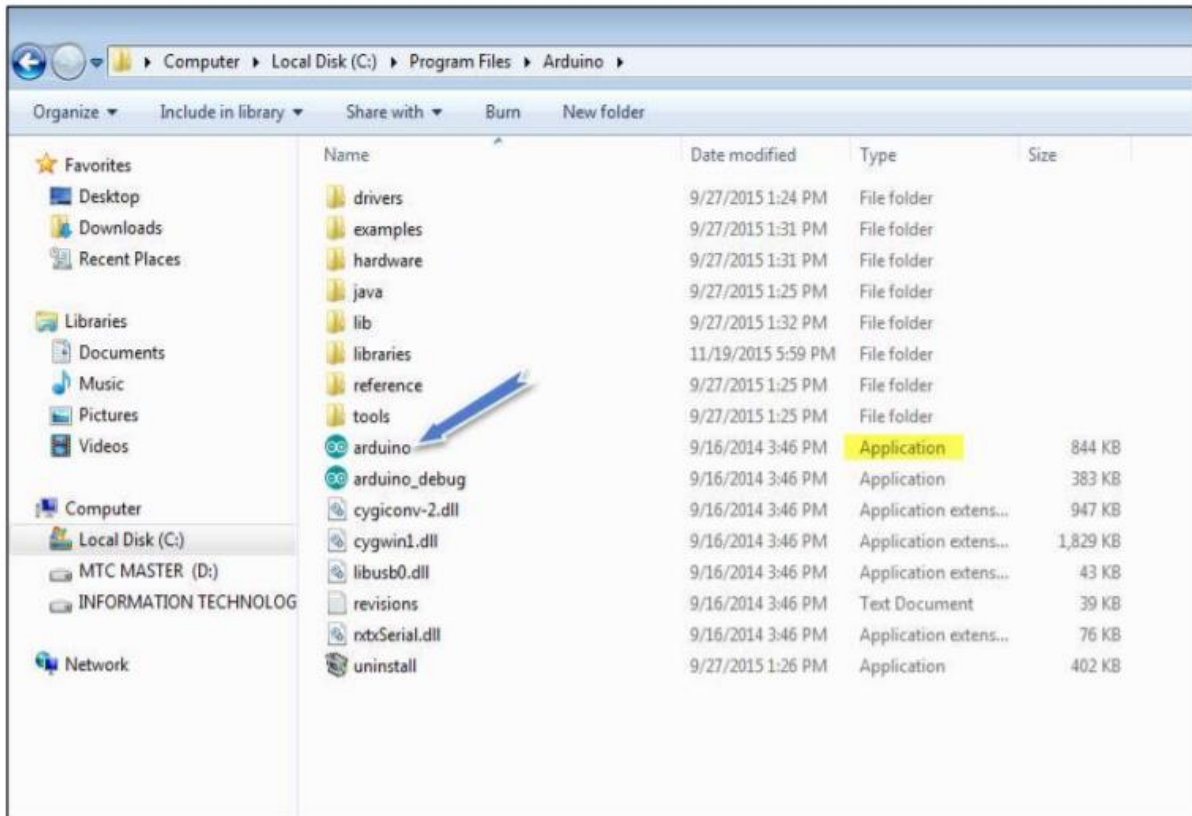
1. Arduino IDE:

Arduino IDE Software. You can get different versions of Arduino IDE from the Download page on the Arduino Official website. You must select your software, which is compatible with your operating system (Windows, IOS, or Linux). After your file download is complete, unzip the file.



Launch Arduino IDE. After your Arduino IDE software is downloaded, you need to unzip the folder. Inside the folder, you can find the application icon with an infinity label (application.exe). Doubleclick the icon to start the IDE.

WOMEN SAFETY NIGHT PATROLLING ROBOT

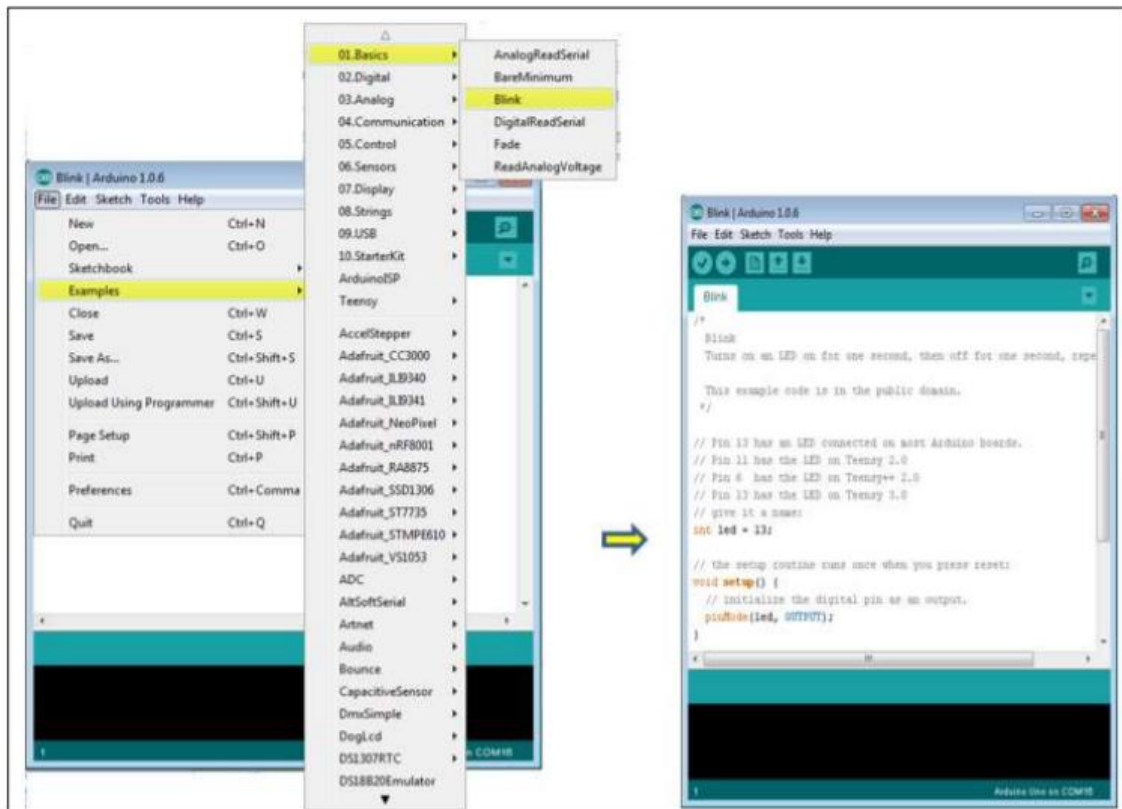
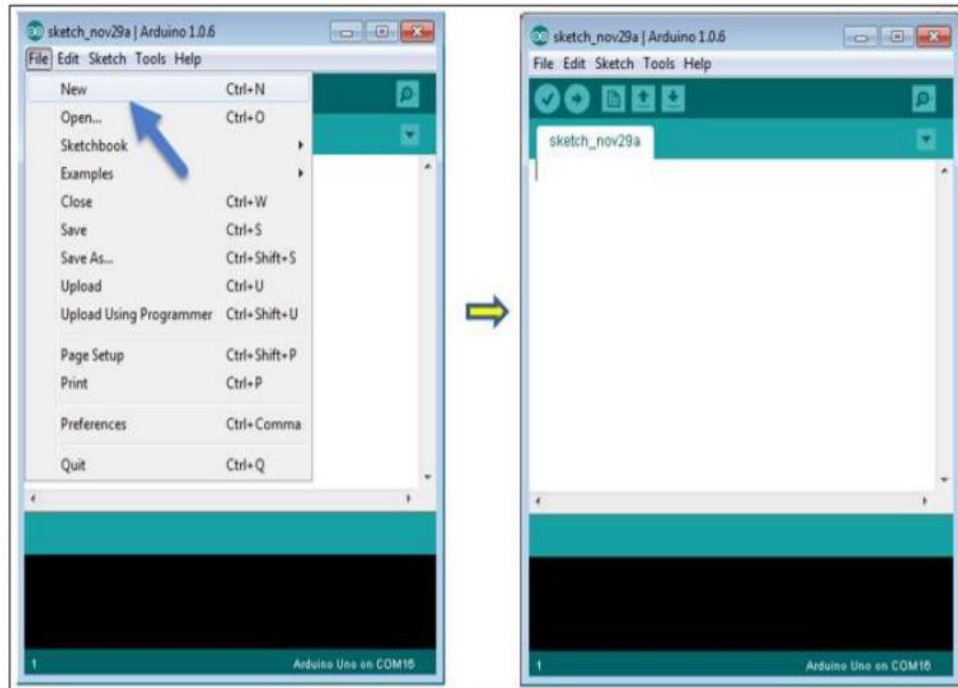


Open your first project. Once the software starts, you have two options:

- Create a new project.
- Open an existing project example.

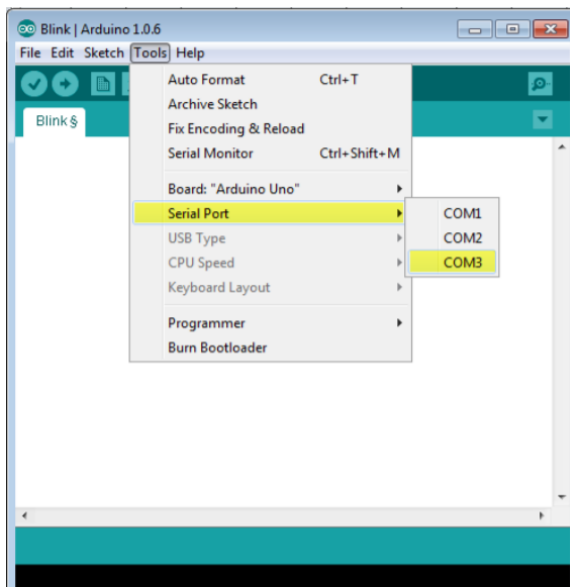
To create a new project, select File --> New

WOMEN SAFETY NIGHT PATROLLING ROBOT



Here, we are selecting just one of the examples with the name Blink. It turns the LED on and off with some time delay. You can select any other example from the list

Select your serial port. Select the serial device of the Arduino board. Go to Tools -> Serial Port menu. This is likely to be COM3 or higher (COM1 and COM2 are usually reserved for hardware serial ports). To find out, you can disconnect your Arduino board and re-open the menu, the entry that disappears should be of the Arduino board. Reconnect the board and select that serial port.



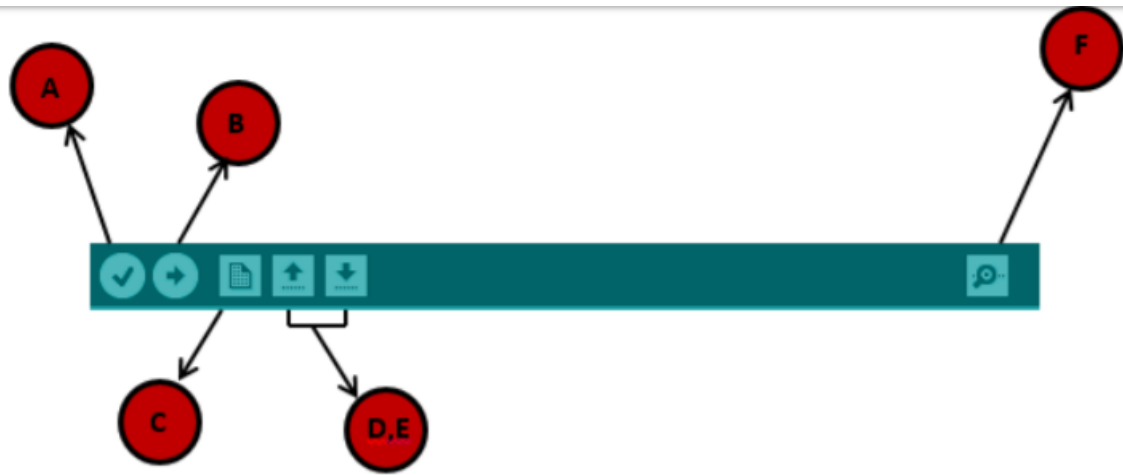
Before explaining how we can upload our program to the board, we must demonstrate the function of each symbol appearing in the Arduino IDE toolbar.

- A- Used to check if there is any compilation error.
- B- Used to upload a program to the Arduino board.
- C- Shortcut used to create a new sketch.
- D- Used to directly open one of the example sketch.
- E- Used to save your sketch.
- F- Serial monitor used to receive serial data from the board and send the serial data to the board.

Now, simply click the "Upload" button in the environment.

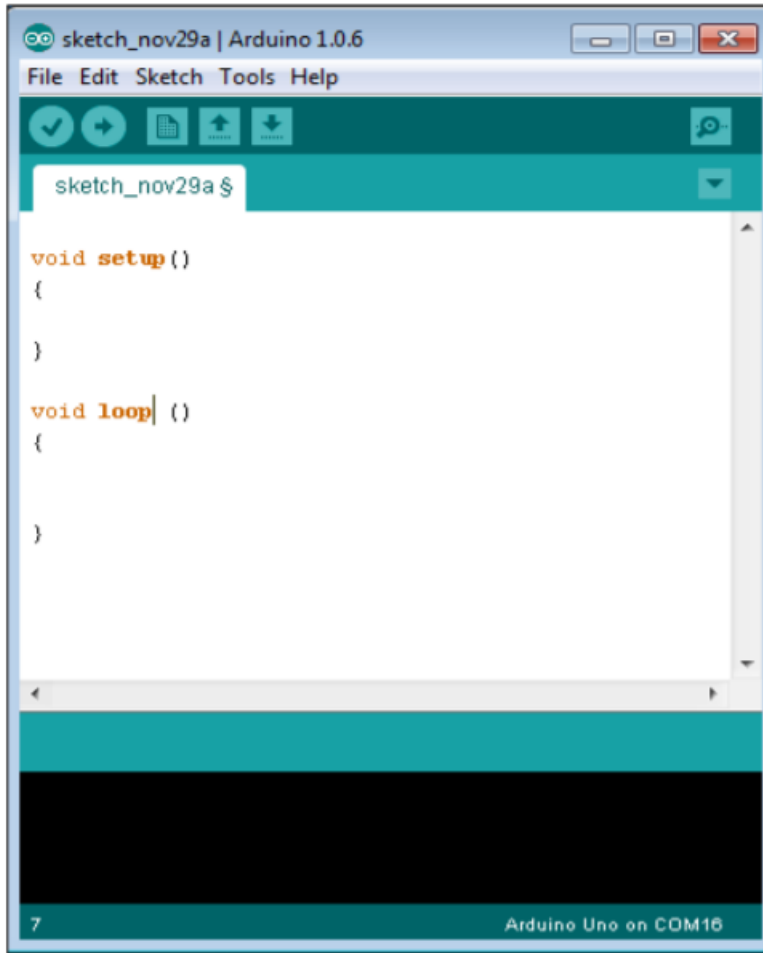
WOMEN SAFETY NIGHT PATROLLING ROBOT

Wait a few seconds; you will see the RX and TX LEDs on the board, flashing. If the upload is successful, the message "Done uploading" will appear in the status bar.



In this chapter, we will study in depth, the Arduino program structure and we will learn more new terminologies used in the Arduino world. The Arduino software is open-source. The source code for the Java environment is released under the GPL and the C/C++ microcontroller libraries are under the LGPL. Sketch: The first new terminology is the Arduino program called “sketch”. Structure Arduino programs can be divided in three main parts: Structure, Values (variables and constants), and Functions. In this tutorial, we will learn about the Arduino software program, step by step, and how we can write the program without any syntax or compilation error. Let us start with the Structure. Software structure consist of two main functions:

- Setup() function
- Loop() function



Data types in C refers to an extensive system used for declaring variables or functions of different types. The type of a variable determines how much space it occupies in the storage and how the bit pattern stored is interpreted. The following table provides all the data types that you will use during Arduino programming.

5.2 CODE

```
//ESP32 Camera Surveillance Car

#include "esp_camera.h"

#include <WiFi.h>

// WARNING!!! Make sure that you have either selected ESP32 Wrover Module,
//          or another board which has PSRAM enabled
// Adafruit ESP32 Feather
// Select camera model

#define CAMERA_MODEL_WROVER_KIT

#define CAMERA_MODEL_M5STACK_PSRAM

#define CAMERA_MODEL_AI_THINKER

const char* ssid = "elegant"; //Enter SSID WIFI Name

const char* password = "smartwork"; //Enter WIFI Password

#if defined(CAMERA_MODEL_WROVER_KIT)

#define PWDN_GPIO_NUM    -1

#define RESET_GPIO_NUM  -1

#define XCLK_GPIO_NUM    21

#define SIOD_GPIO_NUM    26

#define SIOC_GPIO_NUM    27

#define Y9_GPIO_NUM      35
```

WOMEN SAFETY NIGHT PATROLLING ROBOT

```
#define Y8_GPIO_NUM    34
#define Y7_GPIO_NUM    39
#define Y6_GPIO_NUM    36
#define Y5_GPIO_NUM    19
#define Y4_GPIO_NUM    18
#define Y3_GPIO_NUM     5
#define Y2_GPIO_NUM     4
#define VSYNC_GPIO_NUM 25
#define HREF_GPIO_NUM  23
#define PCLK_GPIO_NUM  22

#elif defined(CAMERA_MODEL_AI_THINKER)

#define PWDN_GPIO_NUM  32
#define RESET_GPIO_NUM -1
#define XCLK_GPIO_NUM   0
#define SIOD_GPIO_NUM  26
#define SIOC_GPIO_NUM  27
#define Y9_GPIO_NUM    35
#define Y8_GPIO_NUM    34
#define Y7_GPIO_NUM    39
#define Y6_GPIO_NUM    36
#define Y5_GPIO_NUM    21
#define Y4_GPIO_NUM    19
#define Y3_GPIO_NUM    18
```



```
#define Y2_GPIO_NUM    5

#define VSYNC_GPIO_NUM  25

#define HREF_GPIO_NUM   23

#define PCLK_GPIO_NUM   22


#else

#error "Camera model not selected"

#endif


// GPIO Setting

extern int gpLb = 2; // Left 1

extern int gpLf = 14; // Left 2

extern int gpRb = 15; // Right 1

extern int gpRf = 13; // Right 2

extern int gpLed = 4; // Light

extern String WiFiAddr = "";


void startCameraServer();

void setup() {

    Serial.begin(115200);

    Serial.setDebugOutput(true);

    Serial.println();


    pinMode(gpLb, OUTPUT); //Left Backward
```

```
pinMode(gpLf, OUTPUT); //Left Forward  
pinMode(gpRb, OUTPUT); //Right Forward  
pinMode(gpRf, OUTPUT); //Right Backward  
pinMode(gpLed, OUTPUT); //Light
```

```
//initialize
```

```
digitalWrite(gpLb, LOW);  
digitalWrite(gpLf, LOW);  
digitalWrite(gpRb, LOW);  
digitalWrite(gpRf, LOW);  
digitalWrite(gpLed, LOW);
```

```
camera_config_t config;  
config.ledc_channel = LEDC_CHANNEL_0;  
config.ledc_timer = LEDC_TIMER_0;  
config.pin_d0 = Y2_GPIO_NUM;  
config.pin_d1 = Y3_GPIO_NUM;  
config.pin_d2 = Y4_GPIO_NUM;  
config.pin_d3 = Y5_GPIO_NUM;  
config.pin_d4 = Y6_GPIO_NUM;  
config.pin_d5 = Y7_GPIO_NUM;  
config.pin_d6 = Y8_GPIO_NUM;  
config.pin_d7 = Y9_GPIO_NUM;  
config.pin_xclk = XCLK_GPIO_NUM;
```

```
config.pin_pclk = PCLK_GPIO_NUM;
config.pin_vsync = VSYNC_GPIO_NUM;
config.pin_href = HREF_GPIO_NUM;
config.pin_sscb_sda = SIOD_GPIO_NUM;
config.pin_sscb_scl = SIOC_GPIO_NUM;
config.pin_pwdn = PWDN_GPIO_NUM;
config.pin_reset = RESET_GPIO_NUM;
config.xclk_freq_hz = 20000000;
config.pixel_format = PIXFORMAT_JPEG;
//init with high specs to pre-allocate larger buffers
if(psramFound()){
    config.frame_size = FRAMESIZE_UXGA;
    config.jpeg_quality = 10;
    config.fb_count = 2;
} else {
    config.frame_size = FRAMESIZE_SVGA;
    config.jpeg_quality = 12;
    config.fb_count = 1;
}

// camera init
esp_err_t err = esp_camera_init(&config);
if (err != ESP_OK) {
    Serial.printf("Camera init failed with error 0x%x", err);
```

```
    return;
}

//drop down frame size for higher initial frame rate

sensor_t * s = esp_camera_sensor_get();
s->set_framesize(s, FRAMESIZE_CIF);
WiFi.begin(ssid, password);

while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
}

Serial.println("");
Serial.println("WiFi connected");
startCameraServer();
Serial.print("Camera Ready! Use 'http://");
Serial.print(WiFi.localIP());
WiFiAddr = WiFi.localIP().toString();
Serial.println(" to connect");
}
```

CHAPTER 6

RESULTS AND DISCUSSION

In below fig 9.1 shows the Women Safety Night Patrolling Robot. The robotic vehicle is constructed with a robust chassis, specifically designed to accommodate the ESP32CAM module. The ESP32CAM is a powerful microcontroller equipped with a high-resolution camera, is securely mounted on the robot to capture clear and detailed images. And here L293D Driver is using for to controlling motor wheels.

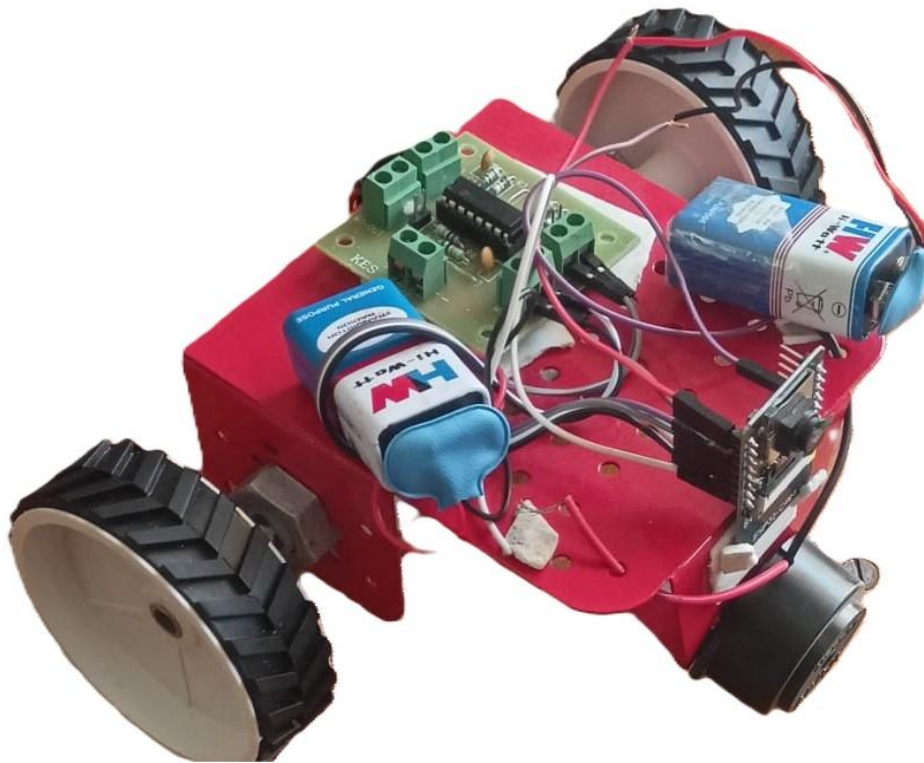


Fig 6.1: Women Safety Night Patrolling Robot

Here we are using two 9v batteries instead of RPS. Batteries are also serving similar functions in providing electrical power, but they are self-contained power source that stores electrical energy in chemical form. Due to instability of the robot here we are using batteries.

Integrating Night Light and A Buzzer into The Women's Safety Patrolling Robot

The buzzer can be activated along with flashing lights to attract attention and signal for help in emergency situations. This can be triggered by the robot itself or by individuals in distress using a panic button. In case of suspicious activity or a safety concern, the robot can activate the buzzer and flash the lights while transmitting a distress signal to nearby authorities or monitoring stations via its communication system. If someone requires assistance, they can activate the buzzer to alert the robot, which will then navigate towards the source of the sound, aided by the night lights, to provide help or guidance. The night lights can c the robot's immediate surroundings, improving visibility for both the robot and individuals nearby, while the buzzer serves as an audible signal of the robot's presence. Overall, the integration of night lights and a buzzer enhances the women's safety patrolling robot's ability to respond to emergencies, deter potential threats, and provide assistance during nighttime patrols.

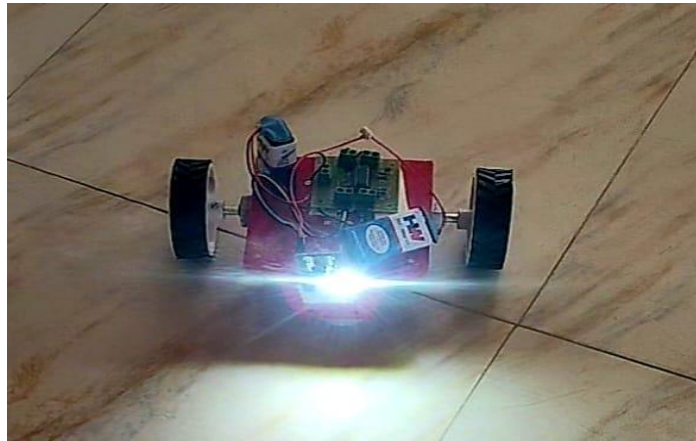


Fig 6.2: Image of Robot When Night light is On

Controlling of Robot Through Mobile Phone and Live Streaming Video

For wireless communication, the robot incorporates Wi-Fi protocols, enabling remote control and monitoring. A unique IP address is assigned to the robot, facilitating seamless control through a mobile device or laptop. It supports protocols like UART/SPI/I2C/PWM/ADC/DAC .

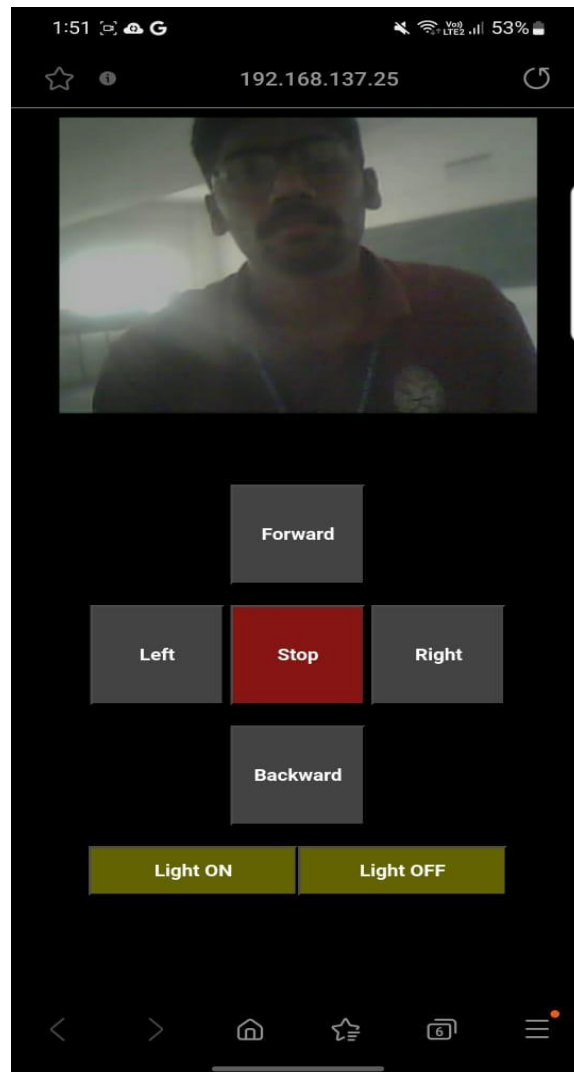


Fig 6.3: Control Panel of Robot (Left, Right, Forward, Backward, Stop, Light On/Off) and Streaming Live Video

In the above figure shows the controls of the robot and streaming live video in the mobile. When we clicking on the forward control the robot moves in forward direction when we click on backward control the robot moves in backward direction and tapping on left button the robot turn to left side and by clicking right the robot turns right side. By clicking on Light Controls, the robot light will turn on and off.

WOMEN SAFETY NIGHT PATROLLING ROBOT

Here the robot and controlling mobile phone must be connected to the same Wi-Fi network then only the controllers will be work in the mobile phone. When the robot connected to a network a unique IP address will be displayed. After that we have to enter that IP address into any internet browser in the same Wi-Fi connected mobile phone. Then the control panel of the robotic vehicle will be visible in the internet browser.



Fig 6.4: Controlling The Robot with Mobile Phone

CHAPTER 7

CONCLUSION

The Women Safety Night Patrolling Robot presents a promising technological solution to address the growing concerns about security in various parts of the world. With its ability to provide continuous surveillance, real-time monitoring, and a remote-controlled interface, the robot contributes significantly to enhancing safety, especially during nighttime hours. The advantages, such as reduced human risk, cost-effectiveness, and community reassurance, underscore the potential of this technology in fostering safer environments. However, like any technological solution, it comes with its set of limitations, including dependency on technology, privacy concerns, and the need for ongoing maintenance and training. These limitations emphasize the importance of a holistic and responsible approach to the deployment of such robots, considering ethical, legal, and social implications.

CHAPTER 8

FUTURE SCOPE

Future of Women Safety Night Patrolling Robot:

The future of Women Safety Night Patrolling Robots holds exciting possibilities for further advancements and improvements. Some key directions for the future development of these robots include:

1. Artificial Intelligence Integration:

Incorporating advanced artificial intelligence (AI) algorithms can enhance the robot's decision-making capabilities, enabling it to respond more effectively to dynamic and complex security situations.

2. Multi-Sensor Integration:

Integrating a variety of sensors, such as infrared sensors, motion detectors, and environmental sensors, can provide a more comprehensive understanding of the surroundings, improving threat detection accuracy.

3. Autonomous Navigation:

Research and development in autonomous navigation systems can enable the robot to navigate and patrol without constant human intervention, optimizing its operational efficiency.

CHAPTER 9

REFERENCES

- [1]. S. Mandapati, S. Pamidi and S. Ambati, "A mobile based women safety application (I Safe Apps)", *IOSR Journal of Computer Engineering*, vol. 17, no. 1, pp. 29-34, 2015.
- [2]. R. S. Yarrabothu and B. Thota, "Abhaya: An Android App for the safety of women", *Annual IEEE India Conference*, pp. 1-4, 2015.
- [3]. S. Murray, "Violence against homeless women: Safety and social policy", *Australian Social Work*, vol. 64, no. 3, pp. 346-360, 2011.
- [4]. V. Hyndavi, N. S. Nikhita and S. Rakesh, "Smart wearable device for women safety using IoT", In 2020 5th International Conference on Communication and Electronics Systems, pp. 459-463, 2020.
- [5]. Sandeep Bhatia, Soniya Verma, Nidhi Singh, Isha Saxena, Ishika verma "IOT and AI Based Women's Safety Night Patrolling Robot" in journal SSRN 13 Jul 2022.
- [6]. J. Ghanem Osman Elhaj Abdalla, 'Implementation of Spy Robot for A Surveillance System using Internet protocol of Raspberry Pi', published in 2017 International conference on recent trends in Electronics information and communication technology.
- [7]. Takato Saito and Yoji Kuroda: 'Mobile robot localization using multiple observations based on place recognition and GPS', published in IEEE International Conference on Robotics and Automation Road detection at night based on a planar reflection model in 2013.
- [8]. Cheng Tang, Quincunxes, Golani Jiang, Yongcheng, 'Road detection at night based on a planar reflection model', published in IEEE International Conference on Information and Automation (ICIA) in 2013.
- [9]. Kirk Mac Tavish, Michael Paton, and Timothy D. Barfoot,: 'Night Rider: Visual Odometry Using Headlights' published in 14th Conference on Computer and Robot Vision in 2017

- [10] .Zhengyuan Han and Tingting Li,: ‘ Research Sound Source Localization and Real-Time FacialExpressionRecognitionforSecurityRobot’,publishedinJournal ofPhysical Conferencein2020
- [11]. Jignesh Patroliya, Haard Mehta, and Hitesh Patel,: ‘Arduino controlled war field spy robot usingnightvisionwirelesscameraandAndroidapplication’,publishedinNirmalaUniversityInternatio
nal ConferenceonEngineeringin 2015
- [12]. Aishwarya K and Baswaraj Gadgay,: ‘IOT Based Smart Multi Application Surveillance Robot’,published in 2ndInternational Conferenceon Inventive Researchin andApplicationsin 2015
- [13]. Eun Som Joen et al.,: ‘Human detection based on the generation of a background image by using farinfraredlightcamera’,Sensors,vol.15,no.3.
- [14]. Dushyant Kumar Singh, Dharmender Singh Kushwaha,: ‘Automatic intruder combat system : Awayto smartboard surviellence’,published in DefenceScienceJournal, vol.67,no1.
- [15]. Dushyant Kumar Singh ,: ‘Review of Human detection techniques in night vision’,published in 2017 International Conference onWirelessCommunications,SignalProcessingandNetworking.
- [16]. Punithavathani, D. Shalini, K. Sujatha, and J. Mark Jain. "Surveillance of anomaly and misuse in critical networks to counter insider threats using computational intelligence." Cluster Computing 18.1 (2015): 435- 451.
- [17] Sujatha, K., and D. Shalini Punithavathani. "Optimized ensemble decision-based multi-focus imagefusion using binary genetic Grey-Wolf optimizer in camera sensor networks." Multimedia Tools and Applications 77.2 (2018): 1735-1759.